HARMILAP SINGH DHALIWAL

Senior Machine Learning Engineer · Applied ML, Retrieval & LLM Systems

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Senior machine learning engineer with 8+ years building and operating applied ML systems across regulated finance, document intelligence, and industrial optimization. Shipped three production AI systems spanning LLM workflows, semantic retrieval, guarded text-to-SQL, and deterministic analytics. Earlier systems delivered $3M in annual savings and improved checkbox-detection accuracy from ~70% to 99.95%.

PROFESSIONAL EXPERIENCE

AI & Data Science Lead, Finance | Royal Bank of Canada

April 2025 - Present

CFO Group, Enterprise Finance · Toronto, ON

Remain ~70% hands-on across architecture, modeling, full-stack implementation, evaluation, and production operations while leading cross-functional delivery.

Production AI Systems

- Built and launched the bank's first production-approved agentic AI platform, owning the core application stack across ETL, document ingestion, PostgreSQL/pgvector dense retrieval, a reviewable authoring workflow, and a single-agent LangGraph orchestrator with bounded MCP tool routines. Piloted in April 2026; launched across the full CFO Group in all geographies in May 2026.
- Conceived, architected, and built the AI/LLM Workforce Analytics Platform from March through its November 2025 production launch, leading cross-functional delivery. It supports authorized headcount, compensation-cost, and open-position questions across ~40,000 cost centres; model stages route intent while deterministic Python retrieves and calculates within the resolved entitlement scope.
- Led a focused two-week v1-to-v2 refactor of the production Financial Peer Benchmarking Platform: dense semantic retrieval ranks candidate metrics; LLM-assisted disambiguation resolves near-duplicate names; reviewed templates and parameter binding constrain text-to-SQL execution.

Technical Leadership

- Set architecture, code-review, and refactoring standards for retrieval, model behavior, and SQL guardrails; manage one Senior AI Scientist, guided the 2025 Amplify cohort, and helped junior contributors ship production changes.
- Own application evaluation and post-release follow-through: prepare LLM-as-judge evidence and extensive human-test results for Model Risk review, backed by custom tracing and logging.

Enterprise Finance Senior Data Scientist | Royal Bank of Canada

September 2022 - April 2025

CFO Group, Enterprise Finance · Toronto, ON

- Built v1 of the Financial Peer Benchmarking Platform end to end, automating text extraction and entity matching across quarterly-changing disclosures from Canada's six largest banks; removed a long-standing peer-analysis bottleneck and earned the 2025 CFO One RBC Team Award.
- Cut commodity-tax return preparation from months to 90 minutes by building a PySpark pipeline supporting ~$600M in allocations from enterprise general-ledger data; earned the CFO Group RBC Quarterly Team Award (Q4 2023).

EARLIER EXPERIENCE

Machine Learning Engineer | Quantiphi Inc.

October 2021 - September 2022

Toronto, ON

- Improved checkbox-detection accuracy from ~70% to 99.95% on Humana claims documents by layering OpenCV pixel-level detection and Random Forest classification over OCR.
- Built document-classification and entity-extraction workflows and deployed custom and AutoML models on Vertex AI; developed supporting data infrastructure with Bigtable and BigQuery.

Data Scientist | Tata Consultancy Services

August 2016 - November 2019

Pune, India

- Delivered $3M in annual savings for a 900 MW generating unit at Kansai Electric's 1,800 MW Maizuru coal-fired station in Japan. As sole data scientist on a three-person R&D team, built the combustion-tuning digital twin end-to-end: ETL across 90+ sensors, 84 regression models, and Particle Swarm Optimization that produced operator-reviewed boiler-tuning recommendations.
- Designed the model-selection framework using k-fold cross-validation, R-squared, RMSE, MAPE, and fold variance; also shipped LSTM forecasting and XGBoost transformer-life prediction models.
- Placed 2nd of 600 in TCS's computer-vision hackathon with a CNN math-notation detector; earned the Innovation Pride Award (2019).

TECHNICAL SKILLS

Programming & Production	Python, SQL, PySpark, ETL, API service design, PostgreSQL/pgvector, Git/GitHub, observability; production delivery through partner-owned OpenShift CI/CD
Machine Learning & Evaluation	PyTorch, TensorFlow, scikit-learn, XGBoost; regression, classification, clustering, time series, CNNs, Transformers; model selection and evaluation
Retrieval & LLM Systems	Document classification and extraction, named entities, embeddings, dense retrieval, semantic ranking, near-duplicate disambiguation, dense RAG, guarded text-to-SQL, LangGraph, MCP, human-in-the-loop workflows, LLM evaluation
Cloud & Data	Google Cloud Platform, Vertex AI, AutoML, BigQuery, Bigtable

EDUCATION

Post Graduate Professional Certificate, Big Data Analytics — Georgian College · 2021

B.Eng., Electronics & Communications Engineering — Thapar University · 2016

AWARDS & RECOGNITION

CFO Conference Award | RBC — 2026

CFO Individual Quarterly Award | RBC — P4 2026